

- Lower the dose
- Take largest dose at bedtime to avoid sedative effects during the day
- Switch to another agent
- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Mild to moderate anxiety: 30–60 mg/day in 3–4 divided doses
- Severe anxiety, anxiety associated with alcohol withdrawal: 45–120 mg/day in 3–4 divided doses

Dosage Forms

- Capsule 10 mg, 15 mg, 30 mg
- Tablet 15 mg

How to Dose

- Titration generally not necessary



Dosing Tips

- Use lowest possible effective dose for the shortest possible period of time (a benzodiazepine-sparing strategy)
- 15 mg tablet contains tartrazine, which may cause allergic reactions in certain patients, particularly those who are sensitive to aspirin
- For interdose symptoms of anxiety, can either increase dose or maintain same total daily dose but divide into more frequent doses
- Can also use an as-needed occasional “top-up” dose for interdose anxiety
- Because anxiety disorders can require higher doses, the risk of dependence may be greater in these patients
- Some severely ill patients may require doses higher than the generally recommended maximum dose
- Frequency of dosing in practice is often greater than predicted from half-life, as duration of biological activity is often shorter than pharmacokinetic terminal half-life

Overdose

- Fatalities can occur; hypotension, tiredness, ataxia, confusion, coma

Long-Term Use

- Risk of dependence, particularly for treatment periods longer than 12 weeks and especially in patients with past or current polysubstance abuse

Habit Forming

- Oxazepam is a Schedule IV drug
- Patients may develop dependence and/or tolerance with long-term use

How to Stop

- Patients with history of seizure may seize upon withdrawal, especially if withdrawal is abrupt
- Taper by 15 mg every 3 days to reduce chances of withdrawal effects
- For difficult-to-taper cases, consider reducing dose much more slowly once reaching 45 mg/day, perhaps by as little as 10 mg per week or less
- For other patients with severe problems discontinuing a benzodiazepine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization
- Be sure to differentiate reemergence of symptoms requiring reinstitution of treatment from withdrawal symptoms
- Benzodiazepine-dependent anxiety patients and insulin-dependent diabetics are not addicted to their medications. When

benzodiazepine-dependent patients stop their medication, disease symptoms can reemerge, disease symptoms can worsen (rebound), and/or withdrawal symptoms can emerge

Pharmacokinetics

- Elimination half-life 3–21 hours
- No active metabolites



Drug Interactions

- Increased depressive effects when taken with other CNS depressants (see Warnings below)



Other Warnings/Precautions

- Boxed warning regarding the increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficulty breathing and death
- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of each drug to the minimum possible while still achieving therapeutic efficacy
- Patients and their caregivers should be warned to seek medical attention if unusual dizziness, lightheadedness, sedation, slowed or difficulty breathing, or unresponsiveness occur

- Dosage changes should be made in collaboration with prescriber
- Use with caution in patients with pulmonary disease; rare reports of death after initiation of benzodiazepines in patients with severe pulmonary impairment
- History of drug or alcohol abuse often creates greater risk for dependency
- Some depressed patients may experience a worsening of suicidal ideation
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)

Do Not Use

- If patient has angle-closure glaucoma
- If there is a proven allergy to oxazepam or any benzodiazepine

Special Populations

Renal Impairment

- Use with caution; oxazepam levels may be increased

Hepatic Impairment

- Use with caution; oxazepam levels may be increased

- Because of its short half-life and inactive metabolites, oxazepam may be a preferred benzodiazepine in some patients with liver disease

Cardiac Impairment

- Benzodiazepines have been used to treat anxiety associated with acute myocardial infarction

Elderly

- Initial 30 mg in 3 divided doses; can be increased to 30–60 mg/day in 3–4 divided doses



Children and Adolescents

- Safety and efficacy not established under age 6
- No clear dosing guidelines for children ages 6–12
- Long-term effects of oxazepam in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR

or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Possible increased risk of birth defects when benzodiazepines are taken during pregnancy
- Because of the potential risks, oxazepam is not generally recommended as treatment for anxiety during pregnancy, especially during first trimester
- Drug should be tapered if discontinued
- Infants whose mothers received a benzodiazepine late in pregnancy may experience withdrawal effects
- Neonatal flaccidity has been reported in infants whose mothers took a benzodiazepine during pregnancy
- Seizures, even mild seizures, may cause harm to the embryo/fetus

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- Effects on infant have been observed and include feeding difficulties, sedation, and weight loss

The Art of Psychopharmacology

Potential Advantages

- Rapid onset of action

Potential Disadvantages

- Euphoria may lead to abuse
- Abuse especially risky in past or present substance abusers

Primary Target Symptoms

- Panic attacks
- Anxiety
- Agitation

Pearls

- Can be a very useful adjunct to SSRIs and SNRIs in the treatment of numerous anxiety disorders
- Not effective for treating psychosis as a monotherapy, but can be used as an adjunct to antipsychotics
- Not effective for treating bipolar disorder as a monotherapy, but can be used as an adjunct to mood stabilizers and antipsychotics
- ✱ Because of its short half-life and inactive metabolites, oxazepam may be preferred over some benzodiazepines for patients with liver disease
- Oxazepam may be preferred over some other benzodiazepines for the treatment of delirium

- Can both cause and treat depression in different patients
- When using to treat insomnia, remember that insomnia may be a symptom of some other primary disorder itself, and thus warrant evaluation for comorbid psychiatric and/or medical conditions
- Though not systematically studied, benzodiazepines have been used effectively to treat catatonia and are the initial recommended treatment

Suggested Reading

Ayd FJ Jr . Oxazepam: update 1989 . *Int Clin Psychopharmacol* 1990 ;**5** :1–15.

Garattini S . Biochemical and pharmacological properties of oxazepam .
Acta Psychiatr Scand Suppl 1978 ;**274** :9 –18.

Greenblatt DJ . Clinical pharmacokinetics of oxazepam and lorazepam .
Clin Pharmacokinet 1981 ;**6** :89 –105.

Oxcarbazepine

Therapeutics

Brands

- Trileptal
- *see index for additional brand names*

Generic?

Yes

Class

- Anticonvulsant, voltage-sensitive sodium channel antagonist

Commonly Prescribed for

(bold for FDA approved)

- **Partial seizures in adults with epilepsy (monotherapy or adjunctive)**
- **Partial seizures in children ages 4–16 with epilepsy (monotherapy or adjunctive)**
- Bipolar disorder

How the Drug Works

- * Acts as a use-dependent blocker of voltage-sensitive sodium channels
- * Interacts with the open channel conformation of voltage-sensitive sodium channels
- * Interacts at a specific site of the alpha pore-forming subunit of voltage-sensitive sodium channels
- Inhibits release of glutamate

How Long Until It Works

- For acute mania, effects should occur within a few weeks
- May take several weeks to months to optimize an effect on mood stabilization
- Should reduce seizures by 2 weeks

If It Works

- The goal of treatment is complete remission of symptoms (e.g., seizures, mania)
- Continue treatment until all symptoms are gone or until improvement is stable and then continue treating indefinitely as long as improvement persists
- Continue treatment indefinitely to avoid recurrence of mania and seizures

If It Doesn't Work (for Bipolar Disorder)

* Many patients have only a partial response where some symptoms are improved but others persist or continue to wax and wane without stabilization of mood

- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider adding psychotherapy
- For bipolar disorder, consider the presence of noncompliance and counsel patient
- Switch to another mood stabilizer with fewer side effects
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)



Best Augmenting Combos for Partial Response or Treatment Resistance

- Oxcarbazepine is itself a second-line augmenting agent for numerous other anticonvulsants, lithium, and atypical antipsychotics in treating bipolar disorder, although its use in bipolar disorder is not yet well studied
- Oxcarbazepine may be a second- or third-line augmenting agent for antipsychotics in treating schizophrenia, although its use in schizophrenia is also not yet well studied

Tests

- Consider monitoring sodium levels because of possibility of hyponatremia, especially during the first 3 months

Side Effects

How Drug Causes Side Effects

- CNS side effects theoretically due to excessive actions at voltage-sensitive sodium channels

Notable Side Effects

- * Sedation (dose-dependent), dizziness (dose-dependent), headache, ataxia (dose-dependent), nystagmus, abnormal gait, confusion, nervousness, fatigue
- * Nausea (dose-dependent), vomiting, abdominal pain, dyspepsia
- Diplopia (dose-dependent), vertigo, abnormal vision
- * Rash



Life-Threatening or Dangerous Side Effects

- Hyponatremia
- Rare activation of suicidal ideation and behavior (suicidality)

Weight Gain



- Occurs in significant minority
- Some patients experience increased appetite

Sedation



- Occurs in significant minority
- Dose-related
- Less than carbamazepine
- More when combined with other anticonvulsants
- Can wear off with time, but may not wear off at high doses

What to Do About Side Effects

- Wait
- Wait
- Wait
- Switch to another agent

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 1200–2400 mg/day

Dosage Forms

- Tablet 150 mg, 300 mg, 600 mg
- Extended-release tablet 150 mg, 300 mg, 600 mg
- Liquid 300 mg/5 mL

How to Dose

- Monotherapy for seizures or bipolar disorder: initial 600 mg/day in 2 doses; increase every 3 days by 300 mg/day; maximum dose generally 2400 mg/day
- Adjunctive: initial 600 mg/day in 2 doses; each week can increase by 600 mg/day; recommended dose 1200 mg/day; maximum dose generally 2400 mg/day
- When converting from adjunctive to monotherapy in the treatment of epilepsy, titrate concomitant drug down over 3–6 weeks while titrating oxcarbazepine up over 2–4 weeks, with an initial daily oxcarbazepine dose of 600 mg divided in 2 doses
- Extended-release: initial 600 mg once per day; increase weekly by 600 mg once per day; recommended maintenance dose 1200–2400 mg once per day



Dosing Tips

* Doses of oxcarbazepine need to be about one-third higher than those of carbamazepine for similar results

- Usually administered as adjunctive medication to other anticonvulsants, lithium, or atypical antipsychotics for bipolar disorder
- Side effects may increase with dose
- Although increased efficacy for seizures is seen at 2400 mg/day compared to 1200 mg/day, CNS side effects may be intolerable at the higher dose
- Liquid formulation can be administered mixed in a glass of water or directly from the oral dosing syringe supplied
- Slow dose titration may delay onset of therapeutic action but enhance tolerability to sedating side effects
- Should titrate slowly in the presence of other sedating agents, such as other anticonvulsants, in order to best tolerate additive sedative side effects
- When converting from immediate-release to extended-release, higher doses may be necessary

Overdose

- No fatalities reported

Long-Term Use

- Safe

- Monitoring of sodium may be required, especially during the first 3 months

Habit Forming

- No

How to Stop

- Taper
- Epilepsy patients may seize upon withdrawal, especially if withdrawal is abrupt
- * Rapid discontinuation may increase the risk of relapse in bipolar disorder
- Discontinuation symptoms uncommon

Pharmacokinetics

- Metabolized in the liver
- Renally excreted
- Inhibits CYP450 2C19
- * Oxcarbazepine is a prodrug for 10-hydroxy carbamazepine
- * This main active metabolite is sometimes called the monohydroxy derivative or MHD, and is also known as licarbazepine
- * Half-life of parent drug is approximately 2 hours; half-life of MHD is approximately 9 hours; thus oxcarbazepine is essentially a prodrug

rapidly converted to its MHD, licarbazepine

- A mild inducer of CYP450 3A4
- Food does not affect absorption



Drug Interactions

- Depressive effects may be increased by other CNS depressants (alcohol, MAOIs, other anticonvulsants, etc.)
- Strong inducers of CYP450 cytochromes (e.g., carbamazepine, phenobarbital, phenytoin, and primidone) can decrease plasma levels of the active metabolite MHD
- Verapamil may decrease plasma levels of the active metabolite MHD
- Oxcarbazepine can decrease plasma levels of hormonal contraceptives and dihydropyridine calcium antagonists
- Oxcarbazepine at doses greater than 1200 mg/day may increase plasma levels of phenytoin, possibly requiring dose reduction of phenytoin



Other Warnings/Precautions

- Because oxcarbazepine has a tricyclic chemical structure, it is not recommended to be taken with MAOIs, including 14 days after MAOIs are stopped; do not start an MAOI until 2 weeks after discontinuing oxcarbazepine

- Because oxcarbazepine can lower plasma levels of hormonal contraceptives, it may also reduce their effectiveness
- May exacerbate angle-closure glaucoma
- May need to restrict fluids and/or monitor sodium because of risk of hyponatremia
- Use cautiously in patients who have demonstrated hypersensitivity to carbamazepine
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately

Do Not Use

- If patient is taking an MAOI
- If there is a proven allergy to any tricyclic compound
- If there is a proven allergy to oxcarbazepine

Special Populations

Renal Impairment

- Oxcarbazepine is renally excreted
- Elimination half-life of active metabolite MHD is increased
- Reduce initial dose by half; may need to use slower titration

Hepatic Impairment

- No dose adjustment recommended for mild to moderate hepatic impairment
- Use with caution in patients with severe impairment

Cardiac Impairment

- No dose adjustment recommended

Elderly

- Older patients may have reduced creatinine clearance and require reduced dosing
- Elderly patients may be more susceptible to adverse effects
- Some patients may tolerate lower doses better



Children and Adolescents

- Approved as adjunctive therapy or monotherapy for partial seizures in children ages 4 and older
- Ages 4–16 (adjunctive): initial 8–10 mg/kg per day or less than 600 mg/day in 2 doses; increase over 2 weeks to 900 mg/day (20–29 kg), 1200 mg/day (29.1–39 kg), or 1800 mg/day (>39 kg)
- When converting from adjunctive to monotherapy, titrate concomitant drug down over 3–6 weeks while titrating oxcarbazepine up by no more than 10 mg/kg per day each week,

with an initial daily oxcarbazepine dose of 8–10 mg/kg per day divided in 2 doses

- Monotherapy: initial 8–10 mg/kg per day in 2 doses (immediate-release) or 1 dose (extended-release); increase every 3 days by 5 mg/kg per day; recommended maintenance dose dependent on weight
- 0–20 kg (600–900 mg/day)
- 21–30 kg (900–1200 mg/day)
- 31–40 kg (900–1500 mg/day)
- 41–45 kg (1200–1500 mg/day)
- 46–55 kg (1200–1800 mg/day)
- 56–65 kg (1200–2100 mg/day)
- over 65 kg (1500–2100 mg)
- Children below age 8 may have increased clearance compared to adults



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
- * Oxcarbazepine is structurally similar to carbamazepine, which is thought to be teratogenic in humans
- * Use during first trimester may raise risk of neural tube defects (e.g., spina bifida) or other congenital anomalies
- Use in women of childbearing potential requires weighing potential benefits to the mother against the risks to the fetus
- * If drug is continued, perform tests to detect birth defects
- * If drug is continued, start on folate 1 mg/day to reduce risk of neural tube defects
- Antiepileptic Drug Pregnancy Registry: 1-888-233-2334, www.aedpregnancyregistry.org
- Taper drug if discontinuing
- * For bipolar patients, oxcarbazepine should generally be discontinued before anticipated pregnancies
- Seizures, even mild seizures, may cause harm to the embryo/fetus
- Recurrent bipolar illness during pregnancy can be quite disruptive
- * For bipolar patients, given the risk of relapse in the postpartum period, some form of mood stabilizer treatment may need to be restarted immediately after delivery if patient is unmedicated during pregnancy
- * Atypical antipsychotics may be preferable to lithium or anticonvulsants such as oxcarbazepine if treatment of bipolar disorder

is required during pregnancy

- Bipolar symptoms may recur or worsen during pregnancy and some form of treatment may be necessary

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- If drug is continued while breast feeding, infant should be monitored for possible adverse effects
- If infant shows signs of irritability or sedation, drug may need to be discontinued
- Bipolar disorder may recur during the postpartum period, particularly if there is a history of prior postpartum episodes of either depression or psychosis
- * Relapse rates may be lower in women who receive prophylactic treatment for postpartum episodes of bipolar disorder
- Atypical antipsychotics and anticonvulsants such as valproate may be safer than oxcarbazepine during the postpartum period when breast feeding

The Art of Psychopharmacology

Potential Advantages

- Treatment-resistant bipolar and psychotic disorders
- Those unable to tolerate carbamazepine but who respond to carbamazepine

Potential Disadvantages

- Patients at risk for hyponatremia

Primary Target Symptoms

- Incidence of seizures
- Severity of seizures
- Unstable mood, especially mania

Pearls

* Some evidence of effectiveness in treating acute mania; included in American Psychiatric Association's bipolar treatment guidelines as an option for acute treatment and maintenance treatment of bipolar disorder

- Some evidence of effectiveness as adjunctive treatment in schizophrenia and schizoaffective disorders
- Oxcarbazepine is the 10-keto analog of carbamazepine, but not a metabolite of carbamazepine
- Less well investigated in bipolar disorder than carbamazepine

* Oxcarbazepine seems to have the same mechanism of therapeutic action as carbamazepine but with fewer side effects

* Specifically, risk of leukopenia, aplastic anemia, agranulocytosis, elevated liver enzymes, or Stevens-Johnson syndrome and serious rash associated with carbamazepine does not seem to be associated with oxcarbazepine

- Skin rash reactions to carbamazepine may resolve in 75% of patients with epilepsy when switched to oxcarbazepine; thus, 25% of patients who experience rash with carbamazepine may also experience it with oxcarbazepine
- Oxcarbazepine has much less prominent actions on CYP 450 enzyme systems than carbamazepine, and thus fewer drug-drug interactions
- Specifically, oxcarbazepine and its active metabolite, the monohydroxy derivative (MHD), cause less enzyme induction of CYP450 3A4 than the structurally related carbamazepine
- The active metabolite MHD, also called licarbazepine, is a racemic mixture of 80% S-MHD (active) and 20% R-MHD (inactive)
- A related compound, APTIOM, is a dibenz[b,f]azepine-5-carboxamide derivative of eslicarbazepine and is also extensively converted to eslicarbazepine
- APTIOM is now approved as an anticonvulsant, but adequate studies have not been conducted on its potential use as a mood stabilizer